

API Security Risk Analysis Report

FUTURE_CS_03

Student Information

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Task: API Security Risk Analysis

Tools Used: Postman, GitHub

1. Introduction

Application Programming Interfaces (APIs) allow software applications to communicate and exchange data. Since APIs often expose sensitive information, they must be designed with proper security controls. This project analyzes the security of a public REST API using Postman to identify potential security risks and recommend best practices.

2. API Description

API Used: JSONPlaceholder

Base URL:

<https://jsonplaceholder.typicode.com>

JSONPlaceholder is a free REST API used for testing and learning API development. It supports common HTTP methods such as GET, POST, PUT, and DELETE.

3. Methodology

The API was tested using Postman with the following HTTP methods:

- GET
- POST
- PUT

- DELETE

The following security aspects were analyzed:

- Authentication
- Authorization
- Input Validation
- Data Exposure
- Rate Limiting

4. API Testing Results

HTTP Method	Endpoint	Result
GET	/users	Successfully retrieved user data
POST	/users	New user request accepted
PUT	/users/1	User update request accepted
DELETE	/users/1	Delete request accepted

5. Security Findings

Security Risk	Impact	Recommendation
No Authentication	High	Implement JWT or OAuth authentication.
No Authorization Checks	High	Restrict access based on user roles and permissions.
Limited Input Validation	Medium	Validate and sanitize all user inputs.
No Rate Limiting	Medium	Apply request rate limiting to prevent abuse.

Security Risk	Impact	Recommendation
Public Data Exposure	Low	Avoid exposing sensitive information in API responses.

6. Recommendations

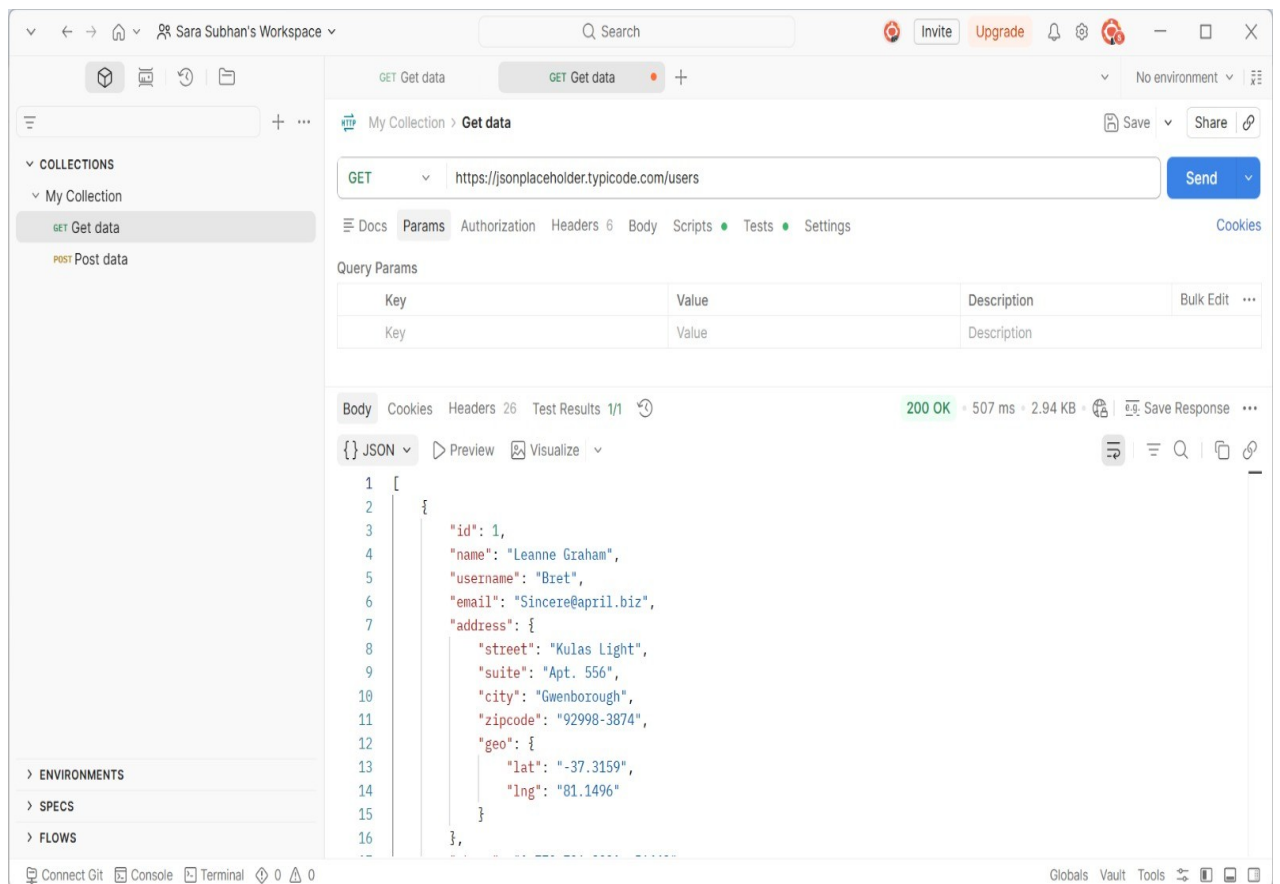
- Implement secure authentication mechanisms.
- Enforce proper authorization for protected resources.
- Validate all client inputs before processing.
- Apply API rate limiting to prevent excessive requests.
- Use HTTPS for secure communication.
- Monitor API logs and security events regularly.

7. Conclusion

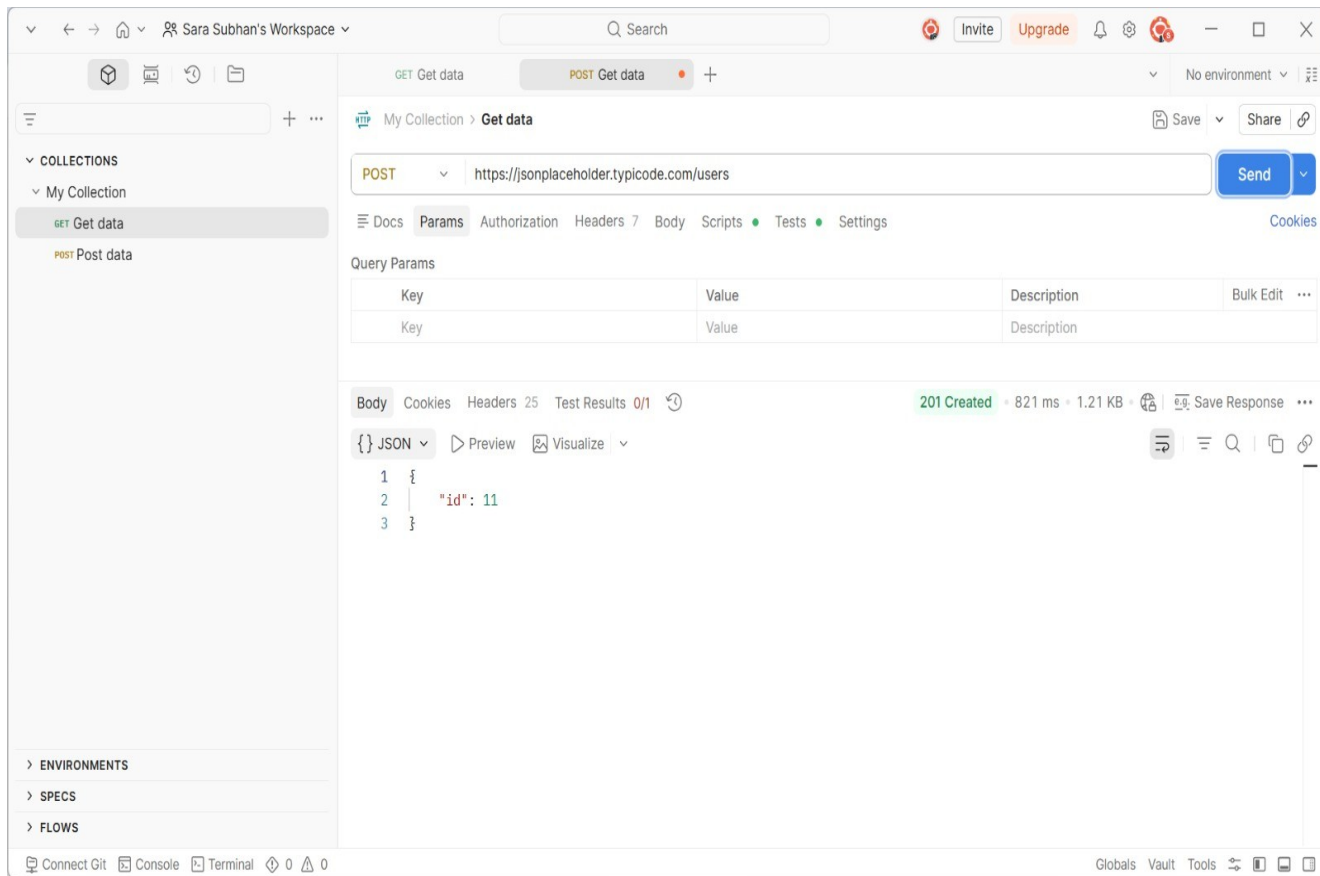
The API testing successfully demonstrated the use of GET, POST, PUT, and DELETE requests using Postman. Although JSONPlaceholder is a testing API and intentionally lacks production security features, this analysis highlights common API security risks and recommends best practices to improve the security of real-world REST APIs.

8. Screenshots

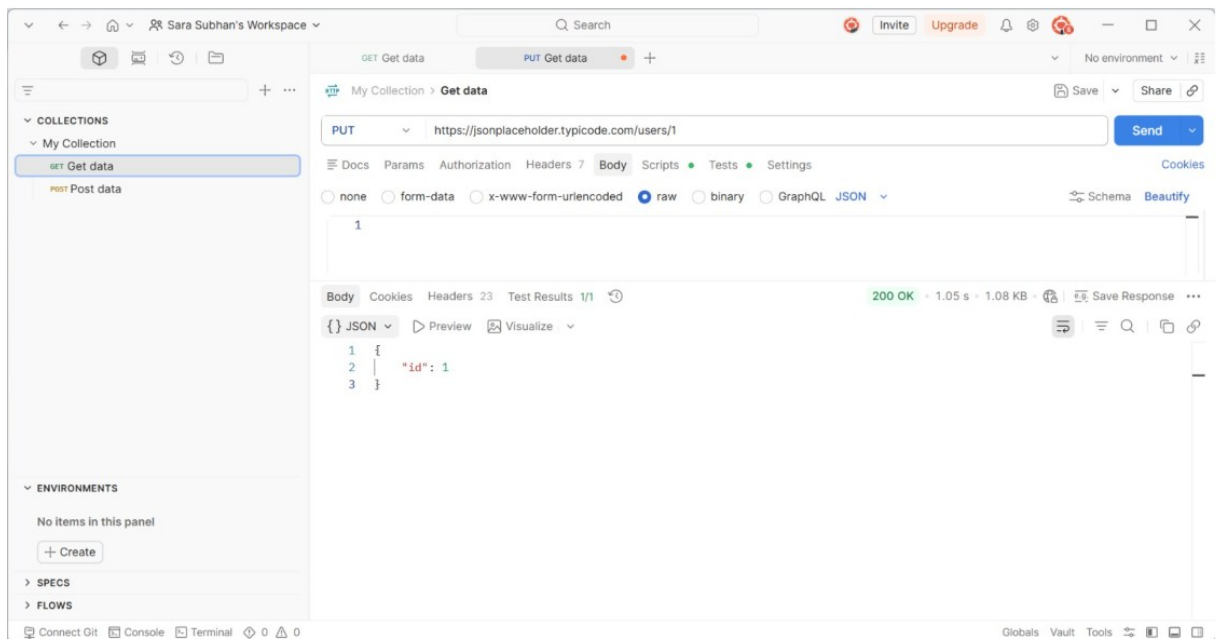
- **Figure 1:** GET Request (get_users.png)



- **Figure 2:** POST Request (post_user.png)



• **Figure 3:** PUT Request (put_user.png)



- **Figure 4:** DELETE Request (delete_user.png)

